

Fumiya Inaba

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Technical Skills

Statistical Analysis & Machine Learning: Hypothesis testing, A/B testing, computer vision (image classification/segmentation), graph neural networks, multiple instance learning, unsupervised clustering, feature extraction/design
Programming: Python, R, SQL - object oriented programming, data visualization, module development,
Frameworks: PyTorch, PyTorch Geometric, Scikit-learn, SciPy, TensorFlow, tidyverse, tidymodels
DevOps: Docker/containerization, git/GitHub,

Experience

PhD (Student) Researcher

Vancouver, BC, Canada

Guillaud & MacAulay Lab @ BC Cancer Research Institute

May 2023–Present

- **[DevOps]** Developed and maintains PyGaulix-Suite, a graphical interface for the annotation, processing (deep learning segmentation), review and analysis of gigapixel whole slide images (WSI) with 4-5 users.
- **[DevOps]** Developed and maintains pygaulix, a Python package serving as the backend for PyGaulix-Suite. Like PyGaulix Suite, has around 4-5 users and used for processing and analysis of histological sections.
- **[Analysis]** Incorporated attention-based segmentation models into existing cell nuclei instance segmentation pipelines; reduced model size by over 80% without a statistically significant drop in DICE score on benchmark test data
- **[Leadership]** Introduced the use of software to facilitate collaboration in and between labs more efficiently using software such as Slack and GitHub, improving collaboration between 4-5 students and 2 labs
- **[Leadership]** Mentored a total of 4 undergraduate research students for completing an 8 month course (PATH 438), on their respective research projects
- **[Leadership]** Acted as a teaching assistant in PATH 305, a digital microscopy course, mentoring a class of 11 students on their image processing project and grading presentations

Computer Vision Research Assistant (Co-op)

Vancouver, BC, Canada

MacAulay Lab @ BC Cancer Research Institute

Sep 2022–May 2023

- Fine-tuned in-house instance segmentation model via active learning; achieving a 15% improvement in DICE score for prostate biopsy images
- Redesigned image processing pipeline (TensorFlow), enabling 2.5x scaling of training dataset capacity

Data Processing Research Assistant (Co-op)

Vancouver, BC, Canada

Guillaud Lab @ BC Cancer Research Institute

Jul 2022–Sep 2022

- Analyzed and classified over 10,000 individual nuclei across 400 images to enhance accuracy of deep learning segmentation model training
- Utilized TensorFlow and CreateML to develop highly accurate deep learning image classification models distinguishing stroma vs epithelium in hematoxylin and eosin-stained tissue, achieving a 90% accuracy rate

UBC Biomolecular Design (BIOMOD)

Vancouver, BC, Canada

Computational Design Team Leader

Oct 2019–Oct 2021

- Pioneered the DNA origami nanostructure design pipelines utilizing software such as caDNA2, MrDNA to design custom structures for two research projects
- Led the design of a DNA nanoscale hinge structure and an angle-locking mechanism using caDNA2 for potential applications in biomolecular diagnostics and nanorobotics
- Managed and trained a team of 7 over three years, supervising the successful designs of multiple nanostructures across two research projects, one of which won first place in the BIOMOD Japan Open competition in 2022

Research Assistant (Work-learn)

Vancouver, BC, Canada

Hallam Lab @ University of British Columbia

Oct 2021–Apr 2022

- Conducted comparative analysis of polyacrylamide gel electrophoresis (PAGE) gels between WT BL21 cells and transformed cells post-IPTG induction to confirm successful expression of NosZ genes in the plasmid
- Cultivated and transformed BL21 E. coli cells with a plasmid vector encoding accessory proteins integral to the NosZ denitrification pathway
- Validated designed plasmid constructs using restriction digests of transformed BL21 E.coli cells

Education

University of British Columbia (UBC)
Doctor of Philosophy – Interdisciplinary Oncology

Vancouver, BC, Canada
Expected May 2028

University of British Columbia (UBC)
Bachelor of Medical Laboratory Sciences

Vancouver, BC, Canada
May 2022

Publications

1. Inaba F, Keyes M, MacAulay C, Guillaud M. Instance scoring via distillation of multiple instance classifiers for interpretable digital pathology. *SPIE Proceedings Digital Library*. 2026. doi:10.1117/12.3087751
2. Alabi N, Nichol A, Inaba F, Carraro A, El-Hallani S, Guillaud M. Large-scale DNA Organization Analysis of Primary Breast Cancers for Prediction of Axillary Lymph Node Metastases. *The Breast*. 2026. doi:10.1016/j.breast.2026.104712
3. Law M, Susham C, Mackay D, et al. Self-Assembly of a Repeatable DNA Nanohinge System Supporting Higher Order Structure Formation. *bioRxiv*. 2023. doi:10.1101/2023.05.26.542516

Awards

Robert F. Wagner All Conference Best Student Paper Award
SPIE Medical Imaging

Vancouver, BC, Canada
Feb 2026

President's Academic Excellence Initiative Award
University of British Columbia (UBC)

Vancouver, BC, Canada
Jan 2025

Canada Graduate Scholarship – Master's (CGS-M)
Natural Sciences and Engineering Research Council of Canada (NSERC)

Vancouver, BC, Canada
May 2024

Interdisciplinary Oncology Program Research Excellence Award
Interdisciplinary Oncology Program, UBC

Vancouver, BC, Canada
Jan 2024

BPOC Graduate Excellence Award
University of British Columbia (UBC)

Vancouver, BC, Canada
Sep 2023

Faculty of Medicine Graduate Award
Interdisciplinary Oncology Program, UBC

Vancouver, BC, Canada
May 2023

Conference Presentations

SPIE Medical Imaging 2026
Oral Presentation

Vancouver, BC, Canada
Feb 2026

Instance scoring via distillation of multiple instance classifiers for interpretable digital pathology

AACR Special Conference: Artificial Intelligence and Machine Learning
Poster Presentation

Montreal, QC, Canada
Jul 2025

Multiple Instance Learning of Large-scale DNA Organization to Characterize Prostate Cancer Aggressiveness

B.I.G Research Day 2025 (BIG25)
Poster Presentation

Vancouver, BC, Canada
Mar 2025

Attention Enhanced Sequential U-Net for Nuclear Segmentation

Quantitative Bio-Imaging Society – Multiplex Image Analysis Workshop
Oral Presentation

San Diego, CA, USA
Sep 2024

Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients

TCAIREM Trainee Rounds	Toronto, ON, Canada
<i>Virtual Oral Presentation</i>	<i>Aug 2024</i>
Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients	
Pathology Day 2024	Vancouver, BC, Canada
<i>Poster Presentation</i>	<i>Jun 2024</i>
Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients	
B.I.G Research Day 2024 (BIG24)	Vancouver, BC, Canada
<i>Poster Presentation</i>	<i>Mar 2024</i>
Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients	
BC Cancer Summit 2023	Vancouver, BC, Canada
<i>Oral Presentation</i>	<i>Nov 2023</i>
Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients	
Canadian Cancer Research Conference 2023	Halifax, NS, Canada
<i>Virtual Poster Presentation</i>	<i>Nov 2023</i>
Large-scale DNA Organization Predicts Aggressive Prostate Cancer in Low and Intermediate Risk Radical Prostatectomy Patients	
BIOMOD Japan Open 2022	Virtual, Japan
<i>Virtual Oral Presentation</i>	<i>May 2022</i>
The Nanohinge: Diagnosing the Problem of Complex Structure Design in Low Carbon Economy	
PATH 438 Directed Studies Presentation Day	Vancouver, BC, Canada
<i>Virtual Oral Presentation</i>	<i>Apr 2022</i>
Optimization of a Methylation-Specific qPCR Assay for Cell-Free Fetal DNA Quantification in Maternal Plasma	

Affiliations

BC Cancer Research Centre, PHSA	Vancouver, BC, Canada
<i>Graduate Student</i>	<i>May 2023–Present</i>
UBC, Department of Pathology and Laboratory Medicine	Vancouver, BC, Canada
<i>Undergraduate/Graduate Student</i>	<i>Sep 2020–Present</i>
UBC Data Science and Health (DASH) Cluster	Vancouver, BC, Canada
<i>Graduate Student/Trainee</i>	<i>Sep 2023</i>